SYNTHETIC AUTOSAR HIGH-LEVEL DESIGN: BRAKE-BY-WIRE (BBW) SYSTEM

DOCUMENT NOTICE: [SYNTHETIC DEMONSTRATION ARTIFACT] Synthetic engineering high-level design specification for TechPulse FY-26 Capstone evaluation. Not an approved OEM production design.

1. System Description and Scope

Document ID: HLD-SYN-BBW-002 | Revision: 1.4 | ASIL Allocation: ASIL D
The Electronic Brake-by-Wire (BBW) system decouples the mechanical brake pedal from electro-hydraulic caliper actuators. Driver braking intent is measured via dual-redundant Hall-effect travel and force sensors.

2. Subsystem Architecture Overview

The architecture incorporates dual microcontrollers running lockstep cores with diverse memory protection units (MPUs). AUTOSAR Classic RTE partitions safety-critical braking loops from comfort functions.

3. Software Components and Port Matrix

Component Name: PedalSensorReader_SWC
Description: Polls primary and secondary brake pedal travel sensors. Performs plausibility checks and cross-channel sensor voting.
Ports:
- **RPort_PedalAngleChA** & **RPort_PedalAngleChB**: ADC hardware abstraction inputs.
- **PPort_ValidatedPedalTravel**: Interface *If_PedalTravel* (Sender-Receiver), provides normalized pedal percentage [0.0 - 100.0%].

Component Name: BrakeDecelArbitrator_SWC
Description: Arbitrates between driver deceleration request, Adaptive Cruise Control (ACC) braking requests, and Autonomous Emergency Braking (AEB) commands.
Ports:
- **RPort_AebBrakeRequest**: Interface *If_ExternalDecelRequest* (from ADAS Gateway).
- **PPort_TargetAxleTorque**: Interface *If_BrakeTorqueCommand* (Client-Server), outputs front/rear target braking torques.

Component Name: WheelBrakeActuator_SWC
Description: Closed-loop PID controller driving brushless motor caliper clamps with slip ratio regulation (ABS mode).

4. Functional Flow: Emergency Braking Sequence

Sequence FF-BBW-02: AEB Deceleration Clamping
1. ADAS radar/camera fusion module issues urgent deceleration request of -8.5 m/s^2 via FlexRay/CAN-FD.
2. *BrakeDecelArbitrator_SWC* verifies driver pedal override threshold.
3. If driver has not applied higher brake pressure, AEB torque priority overrides cruise torque.
4. *WheelBrakeActuator_SWC* commands four wheel calipers with maximum ramp rate within 12 milliseconds.
5. Active wheel speed sensors feedback slip values to modulate hydraulic pressure and prevent lockup.

5. Fail-Safe Strategy & Design Constraints

- **Primary Fail-Safe:** If primary 12V power drops below 9.0V, the secondary auxiliary backup battery immediately engages via solid-state switch.
- **ASIL D Requirement:** Unintended braking hazard prevention (ASIL D); Maximum tolerable transition time to Fail-Silent state is 10 ms.
- **Watchdog Protocol:** External hardware watchdog requires challenge-response token validation every 2 ms from BrakeDecelArbitrator_SWC.